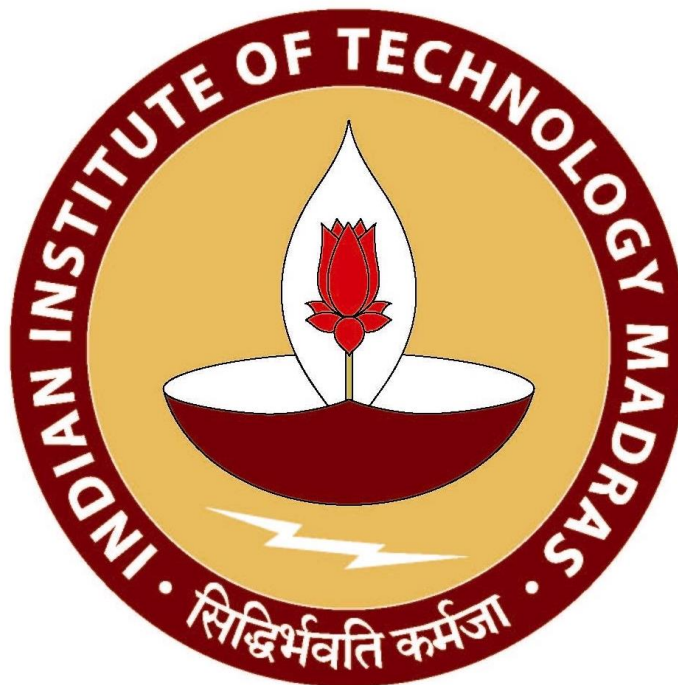


Optimizing Operational Efficiency and Seasonal Profitability: A Data-Driven Strategy for an Air Conditioner Repair Center

A Mid-term report for the BDM capstone Project

Submitted by - Akbar Ali
Roll Number - 23f1002997



IITM Online BS Degree Program,
Indian Institute of Technology, Madras, Chennai
Tamil Nadu, India, 600036

Contents –

1. Executive Summary	2
2. Proof of originality	2
3. Metadata	6
4. Descriptive Statistics	8
5. Detailed Explanation of Analysis	10
6. Results and Findings	11

1. Executive Summary –

Aone Cooling Center, a medium air conditioner repair business, specializes in providing efficient and high-quality repair services. Holding a strong position, still the business faces key challenges regarding procurement cost management and workforce cost optimization. This project is based on original data collected from Aone Cooling Center. The data sources include service records, supplier invoices, and workforce logs maintained by the business owner. A formal authorization letter confirms data authenticity, supported by video documentation and photographic evidence of business operations.

The dataset covers the period from January 2023 to December 2024, encompassing seasonal demand vs cost fluctuations and workforce requirements. The data is structured in spreadsheets compiled from physical records, providing insights into the operational dynamics of the business. The dataset is categorized into two main areas: procurement and workforce. Procurement data consists of purchase details, unit costs, seasonal variations, and pricing trends. The workforce data tracks the number of hired workers, service requests per month, workforce shortages, and excess hiring trends. Descriptive analysis reveals significant fluctuations in spare part costs, especially before the summer season, as well as inefficiencies in workforce, which lead to high labor costs in the off-season and shortages during peak demand periods.

Seasonal demand was analyzed by dividing the dataset into three periods: January to March, April to July, and August to December. To address these challenges, I am analyzing Buying data (Spare Parts invoices) and the historical records of requirement of various spare parts used in different sections of the year in order to get the most efficient and cost-effective buying of spare parts such that they do neither get bought in excess nor in deficiency. Workforce data (number of hired workers at a time) is being analyzed to develop cost-effective hiring time strategies, ensuring an accurate number of workers are employed throughout various parts of the year. Various tools such as trend analysis and predictive modeling are being tested to improve decision-making.

The findings from this analysis highlight key areas for improvement. Spare part prices increase by an average of 18% during peak season, while workforce inefficiencies contribute to higher operational costs and delays in service delivery during high-demand periods. More on, I will be focusing on refining predictive models, validating buying cost-saving strategies, and assessing the effectiveness of implemented changes. By the end of the final report, my goal is to provide actionable recommendations that in turn enhance operational efficiency, improve cost management, and ensure long-term sustainability for Aone Cooling Center in a such a competitive market.

2. Proof of originality –

Primary Data link:

The google sheet link of Primary data which is collected from the business for the analysis of the project is given below.

(Link: [Primary Data Link](#))

Interaction with the Owner, video link:

The google drive link of the in-person interaction meeting with the owner of the business Mr. Asif Khan.

[Interaction Video Link](#)

Images of the Shop:

The photo with the owner and the shop.



The photos of the workshop



A survey link was not required in this situation and hence it is not used.

Letter from the organisation:

This is a letter from the business i.e. Aone Cooling Centre certifying the originality of the data and their permission to use their data for project purposes.

Asif: 89220 34947



Repair of Window A.C., Split A.C. Tower, Casset, Refrigerator, Water Cooler, Water Dispenser, Deep Freezer & Washing Machine, Micro Wave, Geezer etc.

Date: 02-03-2025

To whomsoever it may concern,

This letter serves as certification from Aone Cooling Centre regarding the originality of the data shared with Akbar Ali (23f1002997), a student enrolled in the BS degree program at IIT Madras, for his project of managing a business data.

The data provided by Aone Cooling Centre to Akbar Ali is original and exclusive to our Business. We acknowledge and approve the utilization of this original data by Akbar Ali solely for academic research and analysis purposes related to his project at IIT Madras.

If you require any further confirmation or details regarding the originality of the provided data, please do not hesitate to contact us.

Warm regards,

AONE Cooling Centre

A handwritten signature in black ink, appearing to read "Asif Khan", is written over a horizontal line.

Asif Khan

Owner

3. Metadata –

In order to conduct a thorough analysis of operational efficiency and seasonal profitability at Aone Cooling Center, data was gathered from multiple sources, ensuring accuracy and relevance.

Range of Date –

The range of date of data collected is JAN,2023 TO DEC,2024

Data Sources –

The primary data sources included company records of usage of spare parts while repairing, Bills of cost of spare parts in different parts of the month. Company records provided historical insights into inventory levels, procurement costs, and workforce expenditures. Bills were examined through in order to get to know the market trends of cost of spare parts. Additionally, Company records, including service requests and revenue fluctuations, were analyzed to understand seasonal demand patterns.

Forms of Data –

The data exists in multiple formats. Initially, all information was available as handwritten form manually on paper before being transferred to structured formats that is spreadsheets for numerical analysis. Particularly those related to Usage of Spare parts, was retained in text-based notes. Bills indicating the cost variations of these parts throughout different month of the year, were maintained in physical form then I used it in making the cost variation of Spare parts during different months.

Data Collection Process-

The data for this project was collected from various sources, including company records on spare parts usage during repairs, cost invoices for spare parts. Initially, the data existed in handwritten form but was later digitized into spreadsheets for analysis. The company records provided details on spare parts used in repairs and their procurement costs, while invoices gave insights into market price trends for these parts. Worker Data was manually noted by the owner and then I transferred it excel spreadsheet, was complemented with service requests to assess seasonal demand patterns.

Detailed description about the Procurement Cost Data is given below:

Field Name		Data Type	Description
S.No.		Serial number for unique identification of each spare part	INTEGER
Spare Part		Name of the spare part purchased	STRING
Unit Cost	JAN - MAR	Cost of one unit of the spare part in Jan - Mar 2024	FLOAT

	APR JUL	-	Cost of one unit of the spare part in Apr - Jul 2024	FLOAT
	AUG DEC	-	Cost of one unit of the spare part in Aug - Dec 2024	FLOAT
DEMAND 2024	JAN MAR	-	Number of spare parts required in Jan - Mar 2024	INTEGER
	APR JUL	-	Number of spare parts required in Apr - Jul 2024	INTEGER
	AUG DEC	-	Number of spare parts required in Aug - Dec 2024	INTEGER
DEMAND 2023	JAN MAR	-	Number of spare parts required in Jan - Mar 2023	INTEGER
	APR JUL	-	Number of spare parts required in Apr - Jul 2023	INTEGER
	AUG DEC	-	Number of spare parts required in Aug - Dec 2023	INTEGER
QUANTITY PURCHASED 2024	JAN MAR	-	Number of spare parts purchased in Jan - Mar 2024	INTEGER
	APR JUL	-	Number of spare parts purchased in Apr - Jul 2024	INTEGER
	AUG DEC	-	Number of spare parts purchased in Aug - Dec 2024	INTEGER

Detailed description about the Workforce Management Data is given below:

Field Name	Data Type	Description
S_No	INTEGER	Serial number for unique identification of each record
Month	STRING	The month for which workforce data is recorded
Service Requests	INTEGER	Total number of AC repair requests received that month
Hired Workers	INTEGER	Number of workers employed in that month
Request_handled_per_worker	FLOAT	Number of service requests can be handled per worker in a month
Required Workers	INTEGER	Estimated number of workers needed based on demand
Workers Shortage	INTEGER	Difference when Required Workers > Hired Workers (Required Workers - Hired Workers)
Workers Excess	INTEGER	Difference when Hired Workers > Required Workers (Hired Workers - Required Workers)

4. Descriptive Statistics –

Descriptive statistics, used to describe the dataset using statistical measures such as mean, median, mode, minimum and maximum etc., provide highly valuable analytics on data by giving insights into the distribution and variability of the dataset.

Descriptive statistics on the Workforce Management Data is as follows:

Unit Cost Analysis:

Metric	Jan-Mar	Apr-Jul	Aug-Dec
Count (Total Items)	10	10	10
Mean (Average Price)	136	189	136
Standard Deviation	158.69	181.63	158.69
Minimum Price	30	50	30
25th Percentile (Q1)	37.5	70	37.5
Median (Q2 - 50%)	80	135	80
75th Percentile (Q3)	175	225	175
Maximum Price	550	650	550

Demand Analysis (Comparing 2024 vs. 2023):

Metric	2024 Jan-Mar	2024 Apr-Jul	2024 Aug-Dec	2023 Jan-Mar	2023 Apr-Jul	2023 Aug-Dec
Count (Total Items)	10	10	10	10	10	10
Mean Demand	189	136	6.1	18.3	5.5	7.3
Standard Deviation	181.63	158.69	4.75	9.13	4.28	3.34
Minimum Demand	50	30	0	0	0	0
25th Percentile (Q1)	70	37.5	1.5	12.75	1	6
Median (Q2 - 50%)	135	80	7	18	6.5	7
75th Percentile (Q3)	225	175	9	25.75	8.75	9.75
Maximum Demand	650	550	14	30	11	12

Quantity Purchased (Aug-Dec):

```
count      10.000000
mean       11.000000
std        6.146363
min        0.000000
25%        6.250000
50%       12.500000
75%       15.000000
max       20.000000
```

Descriptive statistics on the Workforce Management Data is as follows:

Metric	Service Requests	Hired Workers	Requests Per Worker	Required Workers	Workers Shortage	Workers Excess
Count (Total Months)	24	24	24	24	24	24
Mean (Average)	76.04	8	10	7.63	1.33	1.75
Standard Deviation	36.08	0	0	3.62	2.33	1.7
Minimum (Lowest Value)	30	8	10	3	0	0
25th Percentile (Q1)	52.75	8	10	5	0	0
Median (Q2 - 50%)	64	8	10	6.5	0	2
75th Percentile (Q3)	91.75	8	10	9.25	1.25	3
Maximum (Highest Value)	146	8	10	15	7	5

Most Frequent (Mode) Month:

```
Service Requests      53.0
Hired Workers         8.0
Request Handled Per Worker  10.0
Required Workers      5.0
Workers Shortage      0.0
Workers Excess        0.0
```

I am also providing the colab notebook for reference of descriptive statistics.
(Link: [colab notebook](#))

5. Detailed Explanation of Analysis –

The detailed analysis of data for this project was conducted systematically to optimize procurement cost management and workforce management using a data-driven approach. The following steps were undertaken –

Data Preprocessing and cleaning

The uncleaned procurement and workforce data were collected from historical usage of spare part records, seller bills, and some are taken by the owner's description. The data was structured in spreadsheets and then imported into Google Colab for analysis.

Data integrity checks were performed widely, including handling missing values by replacing with 0, ensuring correct data types, and validating demands, cost fluctuation workforce(hired) records.

Descriptive Statistics

Descriptive statistics were applied to gain insights into the distribution and variation within the dataset. The main statistical measures were calculated which includes various measures such as mean, standard deviation, minimum, and maximum values etc., which in turn to the better understanding with the trends of the market and distribution and variation of the fields in the dataset.

Inventory Assessment and Procurement Strategy Based on Trends

A trend-based method was employed for procurement optimization by assessing historical demand patterns, and then by reorder point calculations –

Seasonal Demand Evaluation

Demand data from 2023 and 2024 was examined across three seasonal periods: January-March, April-July, and August-December. Identified demand patterns were used to refine procurement planning and prevent overstocking.

Price Variation Assessment

Price fluctuations over time were studied to pinpoint high and low-cost periods. Seasonal price patterns were leveraged to strategically schedule purchases, reducing procurement costs by buying at optimal times.

Spare Parts Demand Forecasting Using Predictive Modeling

To optimize procurement strategies, a Time Series Forecasting method was employed for demand forecasting of spare parts. The ARIMA (AutoRegressive Integrated Moving Average) model was utilized to predict future demand based on historical sales data from previous years. Method:

Data from 2023 and 2024 was analyzed to fit the ARIMA model, which was evaluated for accuracy through residual analysis. This forecasting tool enables the business to predict demand fluctuations, allowing for preemptive adjustments in procurement. This helps avoid stock-outs during high-demand periods and reduces procurement expenses during quieter times.

Hired Workforce Optimization Analysis

The optimization of hiring workforce management was achieved using a Linear Programming (LP) approach, targeting to minimize labor costs while ensuring adequate staffing to meet service demand. This model considered variables such as the number of workers employed and the required number of workers based on the required service capacity for each month. The goal was to determine the most efficient number of workers needed to handle the service demand while controlling costs.

Approach:

The Simplex Method, a widely used optimization technique, was applied to minimize labor costs. Constraints included worker availability, service request demand, and worker productivity. The model was also adjusted to account for seasonal demand changes, ensuring a scalable and cost-effective staffing strategy.

This method was chosen over traditional approaches due to:

This approach provides insights based on facts rather than assumptions. The capability to identify cost fluctuations and seasonal demand spikes allows for optimized procurement and hiring strategies. Getting to know about when to buy spare parts and when to hire or layoff reduces waste, overtime costs, and unnecessary stock accumulation. Unlike static procurement models, this approach uses historical trends and statistical analysis to forecast demand in future and required hiring, making the business more adaptable to market conditions.

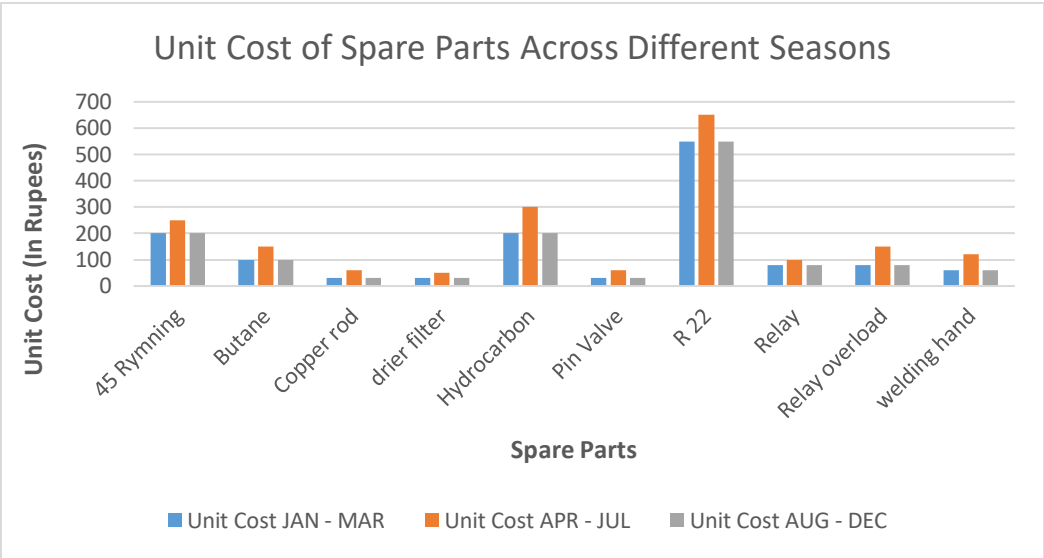
6. Results and Findings –

The analysis of procurement costs factors and workforce hiring requirements and usage, revealed several key trends that impact operational efficiency and profitability at Aone Cooling Center. Following is a detailed breakdown of findings, supported by graphs and interpretations-

a) Unit Cost of spare parts Across Different Seasons –

The price of the spare part increase significantly in Apr-Jul, indicating a seasonal price surge. For e.g., Unit cost of R-22 refrigerant peaked at 650 in Apr-Jul, compared to 550 in

Jan-Mar, showing a +18.2% price hike. Other high-costing spare parts like Hydrocarbon and 45 Rymning also experienced the rise in prices.



b) Required Workers vs. Hired Workers Across Months –

Most of the time worker shortages occurred in peak seasons (Apr-Jul), causing service delays and increased overtime hours and sometimes lead to the faulty services due to workload. In contrast, excess workers in Aug-Feb led to wasted labor costs and hence take part in increasing the costing of services.

